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Response to in class demonstration

EMRs (Electromechanical Relays) and SSRs (Solid State Relays) are two popular types of relays, each with its own pros and cons, as shown in the pros and cons list from our textbook. However, one major issue with EMRs, shown in the in class demonstration, is the potential for arcing in the load and slower response times. While switches (which can be used for AC and DC, as opposed to SSRs as transistors can only be used for DC) are designed to handle high voltage, the voltage arc across the switch can cause the metal to burn and pit over time, leading to long-term reliability issues. Voltage arcing can also contribute to noise and radio frequency interference. The purpose of the demonstration was to illustrate why and under what conditions arcing occurs, and the issues with EMRs. This particular voltage arc was due to the switch turning on at the high points of voltage on the clock wave, and the high sudden input voltage caused an arc that was visible during the demonstration. In the case of EMRs, the arcing happens when the switch opens and closes. In contrast, SSRs do not have this problem, as transistors do not cause arcing. This lack of arcing is a key advantage of solid-state relays.

ELECTRICAL CONTROL DEVICES

Advantages and Disadvantages of Electromechanical and Solid State Relays

Note: Pluses indicate advantages; minuses indicate disadvantages.

GENERAL CHARACTERISTICS	EMR	SSR
1. Arcless switching of the load	-	+
2. Electronic (IC, etc.) compatibility for interfacing	-	+
3. Effects of temperature	+	-
4. Shock and vibration resistance	-	+
5. Immunity to improper functioning because of transients	+	-
6. Radio frequency switching	+	-
7. Zero voltage turn-on	-	+
8. Acoustic noise	-	+
9. Selection of multipole, multithrow switching capability	+	-
10. General cost (It should be noted that the cost of SSRs in comparison with EMRs has been decreasing.)	+	-
11. Ability to stand surge currents	+	-
12. Response time	-	+
13. Voltage drop in load circuit	+	-
14. AC & DC switching with same contacts	+	-
15. Zero current turn-off	-	+
16. Leakage current	+	-
17. Minimum current turn-on	-	+
18. Life expectancy	-	+

FIGURE 2-13
Comparison of EMRs and SSRs

The most common type of timer is the time-delay ON.